Predictive Modeling of Air Quality ($PM_{2.5}$) in Urban Kazakhstan Using Machine Learning Regressors and Meteorological Indicators

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Abstract

Air pollution, particularly fine particulate matter ($PM_{2.5}$), poses a severe environmental and public health challenge in Kazakhstan's urban centers. This study evaluates the efficacy of machine learning algorithms—specifically Linear Regression and Random Forest Regressor—in predicting daily $PM_{2.5}$ concentrations using key meteorological parameters: ambient temperature, relative humidity, and wind speed. Exploratory Data Analysis (EDA) revealed strong negative correlations between $PM_{2.5}$ levels and both wind speed ($r = -0.66$) and temperature ($r = -0.58$), confirming that thermal inversions and stagnant atmospheric conditions exacerbate winter smog episodes. The empirical results demonstrate that Linear Regression achieved a coefficient of determination (R^2) of 0.858 with a Root Mean Squared Error (RMSE) of 13.29 $\mu g/m^3$, slightly outperforming Random Forest ($R^2 = 0.817$, RMSE = 15.14). These findings underscore the viability of lightweight predictive models for early-warning urban air quality monitoring systems.

1 Introduction

Urban regions in Kazakhstan experience critical seasonal degradation in air quality due to topographical basins, coal-based heating, and traffic emissions.

Research Question: To what extent can standard meteorological features (temperature, humidity, wind speed) accurately forecast daily $PM_{2.5}$ concentration spikes?

Hypothesis: Machine learning regressors will effectively map non-linear microclimate interactions, achieving an $R^2 > 0.80$ in predicting particulate concentration.

2 Methods

Data features include independent variables (X): Ambient Temperature (°C), Relative Humidity (%), and Wind Speed (m/s). The dependent variable (y) is $PM_{2.5}$ concentration ($\mu g/m^3$).

Data was split into an 80% training set and a 20% testing set ($N = 365$). Evaluated via two primary algorithms: Linear Regression and Random Forest Regressor ($n = 100$ estimators).

3 Results

Statistical correlations highlighted strong negative relationships:

- Wind speed demonstrated the highest negative correlation with $PM_{2.5}$ ($r = -0.66$).
- Temperature inversely impacted pollution levels ($r = -0.58$).

Table 1: Model Performance Comparison

Algorithm	R^2 **Score**	**RMSE** ($\mu g/m^3$)	**Key Observation**
Linear Regression	**0.858**	**13.29**	Best accuracy and lowest residual error.
Random Forest	0.817	15.14	Robust, but slightly overfitted.

4 Discussion & Conclusion

Meteorological inputs account for $\approx 86\%$ of the variance in local $PM_{2.5}$ levels. Municipalities can leverage such predictive models to issue public health advisories 24–48 hours before predicted stagnation events without deploying costly sensor grids.